

5	(a)	Explain what is meant by <i>coherent</i> light waves.	
			[1]
		Solution: The light waves have a constant phase difference which does not vary with time.	1
	(b)	Coherent light of wavelength 550 nm is incident normally on a double slit of slit separation 2.1 mm, as shown in Fig. 5.1. Both slits in the double slit arrangement have a width of 0.1 mm. A series of bright and dark fringes forms on a screen placed a distance of 2.4 m from the double slit. The screen is parallel to the double slit.	

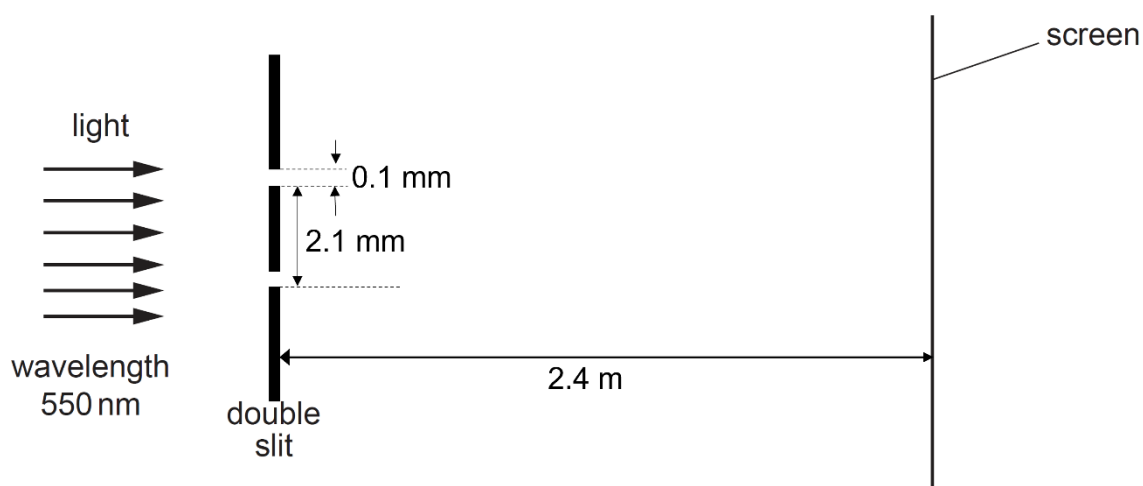


Fig. 5.1 (not to scale)

- (i) Initially, one of the two slits is covered.
- Calculate the width of the central fringe of the single-slit diffraction pattern seen on the screen.
- Give your answer to three significant figures.

fringe width = m [3]

Solution:

For minima of a single slit,

$$\sin \theta_{min} = m \frac{\lambda}{b}$$

For first minima, $m = 1$

$$\sin \theta_{min} = \frac{\lambda}{b} = \frac{550 \times 10^{-9}}{0.1 \times 10^{-3}} = 0.0055$$

$$\tan \theta_{min} = \frac{y}{L}$$

where y is half the width of the central fringe and L is the distance between the slits and the screen

Since θ_{min} is very small, we can use the approximation $\sin \theta \approx \tan \theta$ and then solve for y :

$$\begin{aligned} \sin \theta_{min} &\approx \frac{y}{L} \\ 0.0055 &\approx \frac{y}{2.4} \\ y &= 0.0132 \text{ m} \end{aligned}$$

Fringe width = $0.0132 \times 2 = 0.0264 \text{ m}$ (3sf)

			Solution: red light has longer wavelength (than 550 nm) so distance (between fringes) increases	1
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6	(a)	Use the theory of the particulate nature of electromagnetic radiation to explain why
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